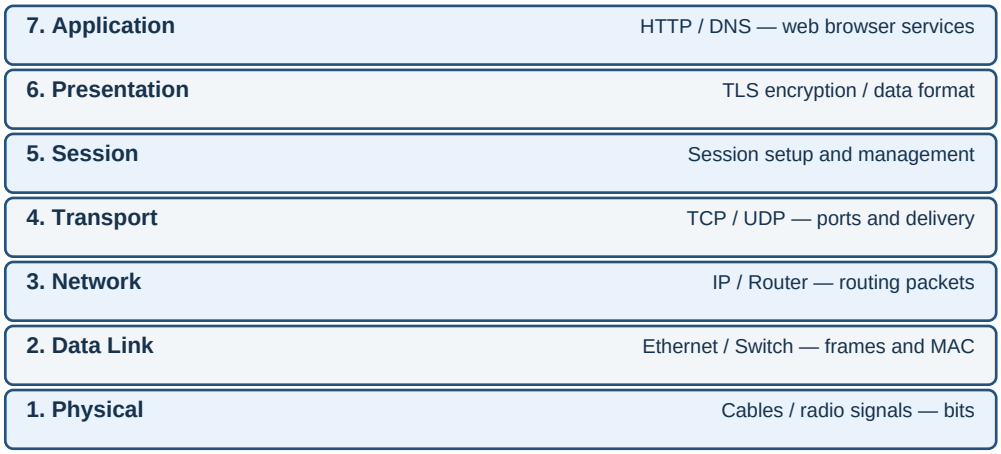
OSI 7-Layer Model Assignment

Name: _____ Class: _____

1. OSI Model Diagram and Examples

The OSI model explains network communication in seven layers. Each layer has a different job.

OSI 7-Layer Model (Top to Bottom)



Layer	Name	Example	Main job
7	Application	HTTP, DNS, web browser	Provides network services to applications.
6	Presentation	TLS encryption, data formats	Encrypts, decrypts, and formats data.
5	Session	Session management	Starts, manages, and ends communication sessions.
4	Transport	TCP, UDP	Delivers data between applications using ports.
3	Network	IP, router	Chooses routes between networks.
2	Data Link	Ethernet, switch, MAC	Moves frames on the local network and checks frame errors.
1	Physical	Cable, fiber, radio signal	Sends raw bits as electrical, light, or radio signals.

2. Match Each Scenario to the OSI Layer

a) Encrypting data before sending: Presentation Layer (Layer 6).

b) Routing packets across networks: Network Layer (Layer 3).

c) Converting data to electrical signals: Physical Layer (Layer 1).

d) Opening a web page in your browser: Application Layer (Layer 7).

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2).

3. OSI Layers When I Send a WhatsApp Message

1. **Application (Layer 7):** I type a message and WhatsApp prepares it to send.

2. **Transport (Layer 4):** Transport protocols help deliver data between the phone and the service.

3. **Network (Layer 3):** IP addressing helps route packets across networks toward the destination.

4. **Data Link (Layer 2):** Wi-Fi or Ethernet sends frames across the local network.

5. **Physical (Layer 1):** Radio waves or cables carry the bits. At the receiving end, the layers work together to deliver the message to the app.

4. Website Does Not Open, but Its IP Address Responds to Ping

If ping to the IP address works, basic IP connectivity is likely working, but the website can still fail at higher layers.

Possible layers:

• **Application (Layer 7):** The web server or website service may be down, or the browser may have a problem.

• **Transport (Layer 4):** TCP port 80 (HTTP) or 443 (HTTPS) may be blocked or not listening.

• **Presentation (Layer 6):** A TLS/SSL certificate or encryption handshake problem may stop HTTPS.

• **Application/DNS:** If the website name does not resolve correctly, DNS may be the problem even when the IP responds.

Reason: Ping usually tests ICMP, not the website's HTTP/HTTPS service. A successful ping does not prove that the web service is working.

Conclusion

The OSI model helps us understand where network problems may occur, from physical signals to applications such as web browsers and messaging apps.